

UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy
DEGREE OF M.Sci. WITH HONOURS
FOURTH-YEAR FINAL EXAMINATION

03 00563

LM GENERAL THEORY OF RELATIVITY

SEMESTER 1 EXAMINATIONS 2021/22

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the *best* two answers.

The approximate allocation of marks to each part of a question is shown in brackets [].

All symbols have their usual meanings.

A formulae sheet will be found at the end of this question paper.

No calculator is permitted in this examination.

Answer **two** questions. If you answer more than two questions, credit will only be given for the best two answers.

1. Consider the motion of a particle of unit mass in the equatorial plane around a neutron star.

(a) Using the Schwarzschild metric given in the attached formulae sheet, write down the corresponding Lagrangian and formulate three conservation laws, including those for energy, $\mathcal{E}$, and angular momentum, J , of the particle. [4]

(b) Use the conservation laws to express the proper radial velocity of the particle in terms of $\mathcal{E}$ and J as follows:

$$\dot{r}^2 = \frac{\mathcal{E}^2}{c^2} - \sigma(r) \left(c^2 + \frac{J^2}{r^2} \right). \quad [2]$$

Now assume that the star has radius $R = 4r_s$, and the particle is in a circular orbit with angular momentum $J = 2cr_s$.

(c) Use the result of part (b) to express

i. the radius of this orbit in terms of r_s and c ; [8]

ii. the energy $\mathcal{E}$ in terms of c^2 . [4]

(d) Determine the orbital period

i. in the frame of the particle; [6]

ii. in the frame of a remote stationary observer. [6]

2. Consider a (1+2)-dimensional spacetime with the following metric:

$$ds^2 = e^{\alpha(r)} dt^2 - e^{-\alpha(r)} dr^2 - r^2 d\phi^2,$$

where r and ϕ are plane polar coordinates, t is a timelike coordinate, $\alpha(r)$ is a differentiable function of r such that $\lim_{r \rightarrow \infty} \alpha(r) = 0$, and units with $c = 1$ are chosen.

(a) Write down the corresponding Lagrangian choosing s as the appropriate affine parameter. [1]

(b) Derive the Euler – Lagrange equations for all the three coordinates. [6]

(c) By writing the derived equations in canonical form, show that the *only* nonvanishing Christoffel symbols for this metric are

$$\Gamma_{01}^0 = \Gamma_{10}^0 = -\Gamma_{11}^1 = \frac{\alpha'(r)}{2}, \quad \Gamma_{00}^1 = \frac{\alpha'(r)}{2} e^{2\alpha(r)}, \quad \Gamma_{22}^1 = -r e^{\alpha(r)}, \quad \Gamma_{21}^2 = \Gamma_{12}^2 = \frac{1}{r}, \quad [4]$$

where the identification $t \equiv x^0$, $r \equiv x^1$ and $\phi \equiv x^2$ is used, and $\alpha' \equiv \frac{d\alpha}{dr}$.

(d) Show that the scalar curvature R of the three-dimensional spacetime with this metric can be expressed via components of the Riemann tensor as follows

$$R = 2 (R^{01}_{01} + R^{02}_{02} + R^{12}_{12}). \quad [2]$$

(e) Express the components of $R^{\mu\nu}_{\alpha\beta}$ featured in part (d) and thus curvature R in terms of the appropriate components of $R^\mu_{\nu\alpha\beta}$. [4]

(f) Calculate the required components of $R^\mu_{\nu\alpha\beta}$ to show that the scalar curvature is given by

$$R = (\alpha' e^\alpha)' + \frac{2\alpha' e^\alpha}{r}. \quad [9]$$

(g) Explain why Einstein's equations in empty space for this metric can only be satisfied by the trivial solution, $\alpha = 0$. What is the physical meaning of this solution? [4]

3. The Friedman – Robertson – Walker (FRW) metric of a homogeneous and isotropic universe is

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 \right]$$

Here r is a dimensionless radial coordinate, $\mathcal{R}(t)$ is a history function, and k is a curvature parameter.

- (a) Explain why it is sufficient to consider only three possible values of k and state these values. [4]

Both $\mathcal{R}(t)$ and k can be found by solving Einstein's equations in the presence of the cosmological term ('vacuum energy'). Consider a model of an **expanding vacuum-dominated** universe where both the radiation and matter contributions are negligible.

- (b) The nonvanishing components of the Riemann tensor for the FRW metric are

$$R^{01}_{01} = R^{02}_{02} = R^{03}_{03} = -\frac{\dot{\mathcal{R}}^2}{\mathcal{R}^2 c^2} \text{ and } R^{12}_{12} = R^{13}_{13} = R^{23}_{23} = -\frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2 c^2}.$$

Use this to show that Einstein's equations for $\mathcal{R}(t)$ and k can be reduced to the following form:

$$\frac{\dot{\mathcal{R}}^2 + kc^2}{\mathcal{R}^2} = \frac{\ddot{\mathcal{R}}}{\mathcal{R}} = \frac{\Lambda c^2}{3} \quad [9]$$

[Hint: You might find the Formulae Sheet attached especially useful for this part and the following part (c).]

- (c) Express $\dot{\mathcal{R}}$ and $\ddot{\mathcal{R}}$ in these equations in terms of the Hubble 'constant' H and 'deceleration' parameter q . [4]

- (d) Given that $q(t_0) = -1/4$ in the current epoch, t_0 , find the curvature parameter k for the model under consideration. [4]

- (e) Using $q(t_0)$ and the initial value of the history function, $\mathcal{R}(t_0) = \mathcal{R}_0$, determine $\mathcal{R}(t)$ for $t > t_0$. [9]

Formulae Sheet

Unless otherwise stated, all indices take values of $\mu = (0, m) \equiv (0, 1, 2, 3)$ in the GR context, or $\mu = (1, \dots, d)$ in a generic d -dimensional space, and the Einstein summation convention is implied

Metric and Lagrangian

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu, \quad \mathcal{L} \equiv g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu, \quad \dot{x}^\mu \equiv u^\mu \equiv \frac{dx^\mu}{ds}, \quad g_{\mu\nu} \text{ is a metric tensor}$$

Here s is any affine parameter along a geodesic line. In GR context, $s \mapsto \tau$ where τ is the proper time

The Schwarzschild metric

$$ds^2 = c^2 \sigma(r) dt^2 - \sigma^{-1}(r) dr^2 - r^2 [d\theta^2 + \sin^2 \theta d\phi^2], \quad \sigma(r) \equiv \left(1 - \frac{r_s}{r}\right), \quad r_s \equiv \frac{2GM}{c^2}$$

Christoffel symbols, $\Gamma_{\alpha\beta}^\mu$

$$\Gamma_{\alpha\beta}^\mu \equiv g^{\mu\nu} \Gamma_{\nu\alpha\beta}, \quad \Gamma_{\nu\alpha\beta} = \frac{1}{2} (g_{\nu\alpha,\beta} + g_{\nu\beta,\alpha} - g_{\alpha\beta,\nu})$$

Covariant derivatives

$$\nabla_\beta \xi^\mu \equiv \xi^\mu_{;\beta} = \xi^\mu_{,\beta} + \Gamma_{\alpha\beta}^\mu \xi^\alpha, \quad \nabla_\beta \xi_\mu \equiv \xi_{\mu;\beta} = \xi_{\mu,\beta} - \Gamma_{\mu\beta}^\alpha \xi_\alpha, \quad \xi^\mu_{,\beta} \equiv \partial_\beta \xi^\mu \equiv \frac{\partial \xi^\mu}{\partial x^\beta}$$

Covariant dot operation

$$\dot{\xi} \equiv \frac{d}{ds} (\xi^\alpha \mathbf{e}_\alpha) = (\nabla_u \xi^\alpha) \mathbf{e}_\alpha \equiv (\xi^\alpha_{,\mu} + \Gamma_{\nu\mu}^\alpha u^\nu) u^\mu \mathbf{e}_\alpha \text{ where } \nabla_u \equiv u^\mu \nabla_\mu$$

Geodesic equations in the Euler–Lagrange form

$$\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) - \frac{\partial \mathcal{L}}{\partial x^\alpha} = 0$$

Geodesic equations in canonical form

$$f^\alpha = \frac{g^{\alpha\beta}}{2} \left[\frac{d}{ds} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\beta} \right) - \frac{\partial \mathcal{L}}{\partial x^\beta} \right] = \ddot{x}^\alpha + \Gamma_{\mu\nu}^\alpha \dot{x}^\mu \dot{x}^\nu = 0 \iff \dot{u}^\alpha + \Gamma_{\mu\nu}^\alpha u^\mu u^\nu = 0 \iff \nabla_u u^\alpha = \nabla_u^2 x^\alpha = 0$$

The Riemann, $R_{\nu\alpha\beta}^\mu$

$$R_{\nu\alpha\beta}^\mu \xi^\nu = \xi^\mu_{;\alpha\beta} - \xi^\mu_{;\beta\alpha}, \quad R_{\nu\alpha\beta}^\mu = -\Gamma_{\nu\alpha,\beta}^\mu + \Gamma_{\nu\beta,\alpha}^\mu - \Gamma_{\nu\alpha}^\sigma \Gamma_{\beta\sigma}^\mu + \Gamma_{\nu\beta}^\sigma \Gamma_{\alpha\sigma}^\mu$$

The Riemann in locally (pseudo)Cartesian frame (LCF)

$$g_{\mu\nu} R^\mu{}_{\alpha\beta\gamma} = R_{\nu\alpha\beta\gamma} = g_{\nu\gamma,\alpha\beta} - g_{\nu\beta,\alpha\gamma} + g_{\alpha\beta,\nu\gamma} - g_{\alpha\gamma,\nu\beta}$$

Symmetries of the Riemann tensor

$$R^\alpha{}_{\beta\mu\nu} + R^\alpha{}_{\mu\nu\beta} + R^\alpha{}_{\nu\beta\mu} = 0, \quad R^\alpha{}_{\beta\mu\nu} = -R^\alpha{}_{\beta\nu\mu}, \quad R_{\alpha\beta\mu\nu} = R_{\mu\nu\alpha\beta}$$

Geodesic deviation

$$f^\alpha = \nabla_u^2 x^\alpha(\tau; \eta_0 + d\eta) = R^\alpha{}_{\beta\mu\nu} u^\beta u^\mu \xi^\nu, \quad u^\mu = \frac{dx^\mu(\tau; \eta_0)}{d\tau}, \quad \xi^\nu = \frac{dx^\nu(\tau; \eta)}{d\eta} d\eta.$$

The Ricci, $R_{\mu\nu}$, and Einstein, $G_{\mu\nu}$, tensors and scalar curvature, R

$$R_{\mu\nu} \equiv R^\alpha{}_{\mu\alpha\nu}, \quad G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R, \quad R = g^{\mu\nu} R_{\mu\nu}.$$

Einstein's equations

$$G_{\mu\nu} = \kappa T_{\mu\nu} \iff R_{\mu\nu} = \kappa \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T^\alpha{}_\alpha \right); \quad \kappa \equiv \frac{8\pi G}{c^4}, \quad T_{\mu\nu} \text{ is a matter tensor.}$$

In the cosmological context, $\Lambda g_{\mu\nu}$ should be added to the r.h.s., where $\Lambda > 0$ is the cosmological constant.

The matter tensor in the perfect fluid model of the Universe

$$T^\mu{}_\nu = (\rho + P/c^2) u^\mu u_\nu - P \delta^\mu{}_\nu.$$

Here ρ is the matter density and P is the radiation pressure.

The Friedmann–Robertson–Walker metric

$$ds^2 = c^2 dt^2 - \mathcal{R}^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2\theta d\phi^2) \right]$$

Here $k = 1, 0$, or -1 is the curvature parameter, and $\mathcal{R}(t)$ is the history function.

The Hubble ‘constant’ H and ‘deceleration’ parameter q

$$H \equiv \frac{\dot{\mathcal{R}}}{\mathcal{R}}, \quad q \equiv -\frac{\ddot{\mathcal{R}}\mathcal{R}}{\dot{\mathcal{R}}^2}, \quad \text{where} \quad \dot{\mathcal{R}} \equiv \frac{d\mathcal{R}}{dt}, \quad \ddot{\mathcal{R}} \equiv \frac{d^2\mathcal{R}}{dt^2}.$$

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches must be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.